Software Design Specification (SDS)

For

Optical Service Web System

HNDIT – 4052

Programming Individual Project



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1. Introduction

1.1 Purpose

This Software Design Specification (SDS) document describes the complete system design of the Optical Service Management System (OSMS). It transforms the requirements defined in the Software Requirements Specification (SRS) into detailed architectural and technical designs including system architecture, database structure, interface design, security mechanisms, and UML diagrams. This document is intended for developers, designers, testers, project supervisors, and system administrators.

1.2 Scope

The Optical Service Management System (OSMS) is a web-based platform developed for optical shops and eye care centers in Sri Lanka. The system digitizes optical service management by handling customer registrations, eye test appointments, prescriptions, product management, online ordering, and report generation.

The system includes:

- Customer Portal
- Appointment Management
- Prescription Management
- Product Management
- Inventory Management
- Online Ordering System
- Payment Management
- Admin Dashboard
- Reporting System

The platform automates optical shop operations, reduces manual work, improves customer service, and enhances business efficiency.

1.3 Definitions and Acronyms

Term	Description
OSMS	Optical Service Management System
UI	User Interface
API	Application Programming Interface
DBMS	Database Management System
Admin	System Administrator
CRUD	Create, Read, Update, Delete
ERD	Entity Relationship Diagram
JWT	JSON Web Token

1.4 References

- IEEE Software Documentation Standards
- React Documentation
- Node.js Documentation
- Express.js Documentation
- MySQL Documentation
- Bootstrap Documentation

2. System Overview

2.1 System Description

The Optical Service Management System is a standalone web application designed to streamline optical shop operations digitally. The platform supports multiple user roles including Customer, Staff Member, and Administrator.

Core subsystems include:

- Appointment Booking System
- Optical Product Marketplace
- Prescription Management
- Inventory Tracking
- Payment Processing
- Report Generation
- User Management

The system is developed using the Waterfall SDLC model to ensure proper planning, development, testing, and deployment.

2.2 System Context

The system interacts with:

- ✓ Customers
- ✓ Optical Shop Staff
- ✓ Administrators
- ✓ Payment Gateway
- ✓ Database Server

User Roles

- ✓ Guest User
- ✓ Registered Customer
- ✓ Staff Member
- ✓ Administrator

2.3 System Architecture

The system follows a 3-Tier Architecture.

Presentation Layer

- React.js frontend
- HTML5
- CSS3
- Bootstrap

Business Logic Layer

- Node.js backend
- Express.js REST APIs

Data Layer

- MySQL database

3. System Requirements

3.1 Functional Requirements

FR ID	Feature	Description	Priority
FR-1	User Registration	Customers create accounts	High
FR-2	Appointment Booking	Customers book eye test appointments	Critical
FR-3	Prescription Management	Staff upload and manage prescriptions	High
FR-4	Product Management	Admin manages glasses and lenses	High
FR-5	Inventory Management	Track product stock levels	High
FR-6	Online Ordering	Customers purchase products online	Critical
FR-7	Payment System	Secure online payments	Critical
FR-8	Report Generation	Generate sales and appointment reports	Medium
FR-9	Notifications	Appointment and order notifications	Medium

3.2 Non-Functional Requirements

Attribute	Specification
Performance	Page load < 3 seconds
Security	JWT authentication and encrypted passwords
Reliability	99% system availability

Usability	Responsive and user-friendly interface
Scalability	Modular architecture for future expansion
Maintainability	Well-documented modular code

4. Design Constraints

4.1 Hardware Constraints

- ✓ Minimum 4GB RAM
- ✓ Dual-core processor
- ✓ Stable internet connection
- ✓ Server storage minimum 100GB

4.2 Software Constraints

- ✓ Operating System: Windows/Linux
- ✓ Browser Support: Chrome, Firefox, Edge
- ✓ Database: MySQL
- ✓ Web Server: XAMPP/Apache

5. Architectural Design

5.1 System Architecture Diagram

The system architecture consists of:

- Client Devices
- Web Application Server
- Database Server

The architecture follows the 3-tier architecture pattern.

5.2 Component Description

Presentation Layer

Handles all user interactions through web interfaces.

Application Layer

Processes business logic including:

- Authentication
- Appointment scheduling
- Order management
- Prescription processing

Database Layer

Stores persistent system data including:

- Customer details
- Prescriptions
- Orders
- Products
- Payments

5.3 Deployment Diagram

The deployment diagram includes:

- User Devices
- Web Browser
- Application Server
- MySQL Database Server

6. Data Design

6.1 Entity Relationship Diagram (ERD)

Main Entities

- USER
- APPOINTMENT
- PRESCRIPTION
- PRODUCT
- ORDER
- PAYMENT
- INVENTORY

Relationships

- ✓ One customer can book many appointments
- ✓ One customer can place many orders
- ✓ One prescription belongs to one customer
- ✓ One admin manages many products

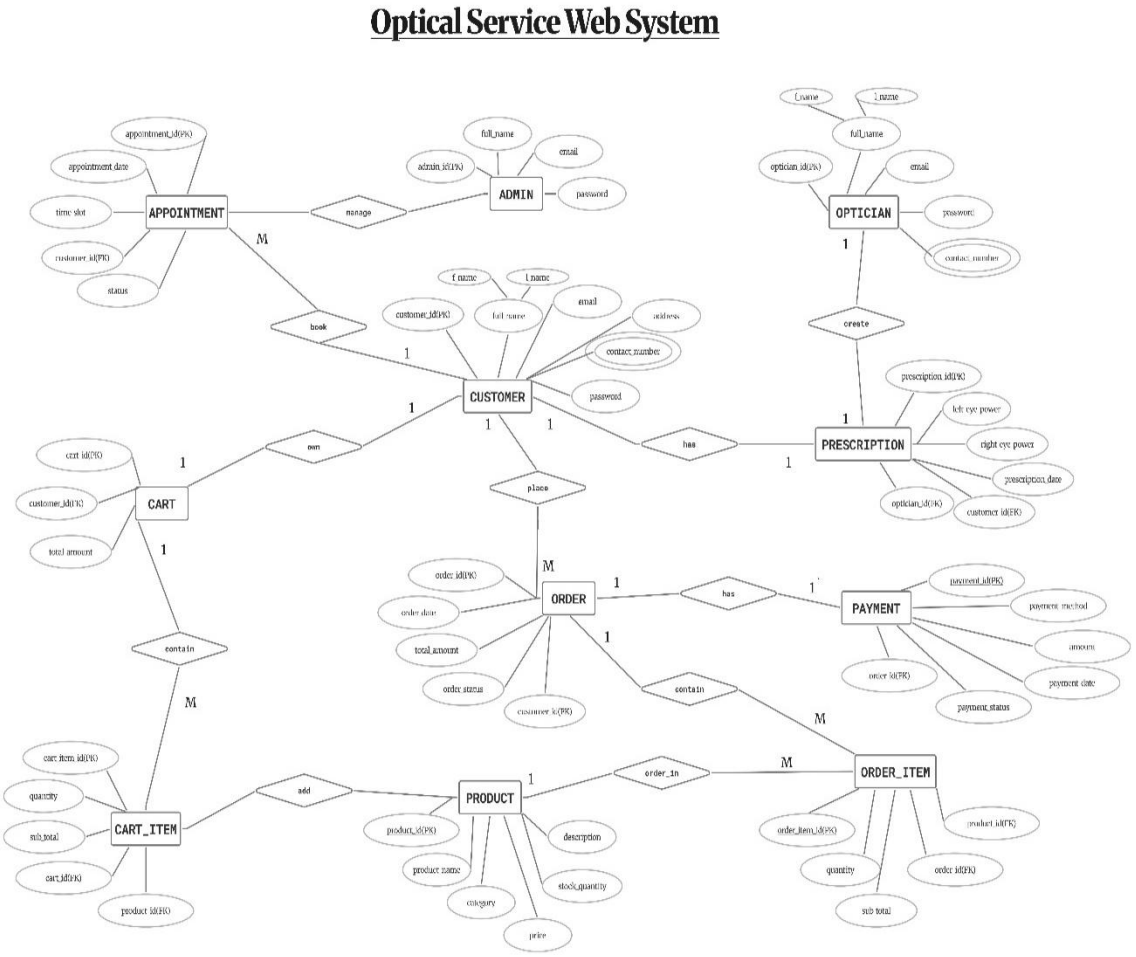


Figure 1- ER Diagram

6.2 Database Schema

USER Table

Column Name	Data Type	Description
user_id	INT	Primary Key
name	VARCHAR(100)	User full name
email	VARCHAR(100)	User email
password	VARCHAR(255)	Encrypted password
role	ENUM	Admin/Staff/Customer
phone	VARCHAR(15)	Contact number

APPOINTMENT Table

Column Name	**Data Type**	**Description**
appointment_id	INT	Primary Key
customer_id appoitment_date	INT DATE	Foreign Key Appointment date
status	ENUM	Pending/Completed

PRODUCT Table

Column Name	Data Type	Description
product_id	INT	Primary Key
product_name	VARCHAR(100)	Product name
category	VARCHAR(50)	Glasses/Lenses
price	DECIMAL(10,2)	Product price
stock_qty	INT	Available stock

ORDER Table

Column Name	**Data Type**	**Description**
order_id	INT	Primary Key
customer_id	INT	Foreign Key
total_amount	DECIMAL(10,2)	Total order value
order_date	TIMESTAMP	Order timestamp

6.3 Data Dictionary

Field Name	**Type**	**Description**
user_id	Integer	Unique user ID
appointment_date	Date	Scheduled appointment
product_name	Varchar	Optical product name
total_amount	Decimal	Total payment amount

7. Component Design

7.1 Use Case Diagram

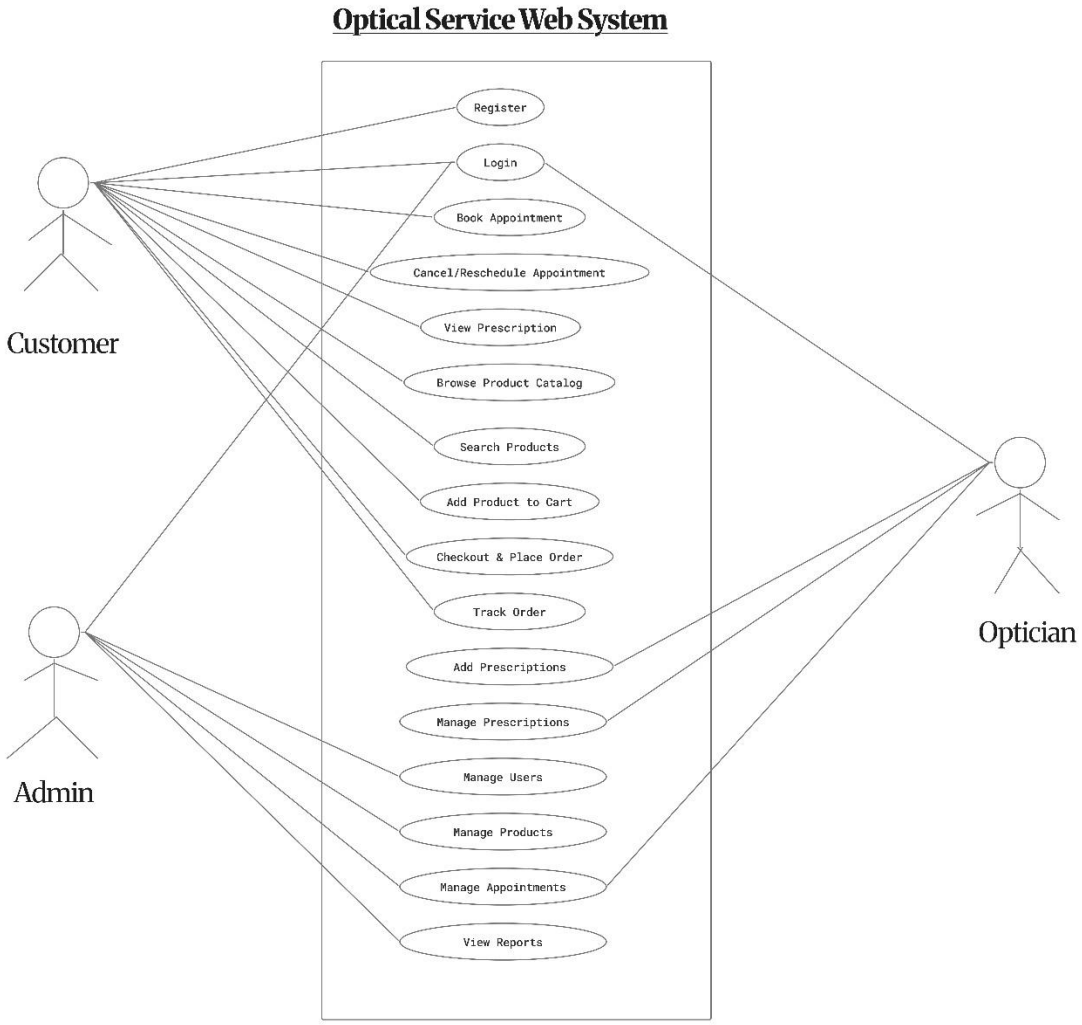


Figure 2- Use case Diagram

7.2 Use Case Description

1. User Registration

The User Registration use case describes the process where a new customer creates an account in the Optical Service Management System. The customer enters personal details such as name, email address, phone number, and password through the registration form. The system validates the entered information and checks whether the email already exists in the

database. If the information is valid, the system creates a new customer account successfully and stores the user data securely in the database. This use case helps customers access system services such as appointment booking and online product purchasing.

2. User Login

The User Login use case explains how registered users access the Optical Service Management System using their email and password. The user enters login credentials, and the system verifies the information with the database. If the credentials are correct, the system grants access and redirects the user to the appropriate dashboard based on their role such as Customer, Staff, or Administrator. If invalid credentials are entered, the system displays an error message and requests the user to try again.

3. Book Appointment

The Book Appointment use case allows customers to schedule eye examination appointments online through the system. The customer selects a preferred date and time from the available appointment slots and confirms the booking request. The system validates the selected schedule and stores the appointment information in the database. After successful booking, the customer receives a confirmation message. This use case improves appointment management efficiency and reduces manual scheduling work.

4. Upload Prescription

The Upload Prescription use case describes the process where staff members enter and manage customer eye prescription details after completing an eye examination. The staff selects the relevant customer profile, enters prescription information such as lens power and eye measurements, and uploads the data into the system. The system validates and stores the prescription securely in the database for future access and product recommendations.

5. Manage Products

The Manage Products use case explains how administrators handle optical product management within the system. The administrator can add new products, update product information, manage pricing, update stock quantities, and remove unavailable products. The system stores all updated product details in the database and ensures inventory information remains accurate for customers and staff members.

6. Purchase Products

The Purchase Products use case allows customers to browse and purchase optical products such as glasses, frames, and contact lenses through the online platform. Customers can search products, add selected items to the shopping cart, and proceed to checkout. The system calculates the total payment amount, processes the order, and stores transaction information in the database. Customers receive an order confirmation after successful purchase completion.

7. Process Payment

The Process Payment use case describes how the system handles online payment transactions for product orders and appointments. The customer selects a payment method and enters payment details. The system securely validates the payment information through the payment gateway and confirms whether the transaction is successful. Once the payment is completed, the system updates the order status and generates a payment confirmation message.

8. Generate Reports

The Generate Reports use case allows administrators to generate system reports related to appointments, sales, inventory, and customer activities. The administrator selects the required report type and specifies report criteria such as date range or category. The system retrieves the relevant data from the database and generates a detailed report that can be viewed, downloaded, or printed for business analysis and decision-making purposes.

9. Manage Inventory

The Manage Inventory use case explains how administrators and staff members monitor and update product stock levels within the optical shop. Users can check available quantities, update inventory records when products are sold or restocked, and receive low-stock alerts. The system ensures accurate inventory management and helps prevent product shortages.

10. Manage Appointments

The Manage Appointments use case describes how staff members and administrators manage customer appointment schedules. Authorized users can view appointment details, confirm bookings, update appointment statuses, reschedule appointments, or cancel appointments when necessary. The system updates the database automatically and ensures accurate appointment tracking and scheduling.

8. Interface Design

8.1 User Interface Design

The system uses a responsive and modern interface.

Main Interfaces

1. Home Page

The Home Page is the main interface of the Optical Service Management System. It provides an overview of the system, featured optical products, services, navigation menus, and quick access to appointments and product categories. The page is designed to attract users and provide easy navigation throughout the system.

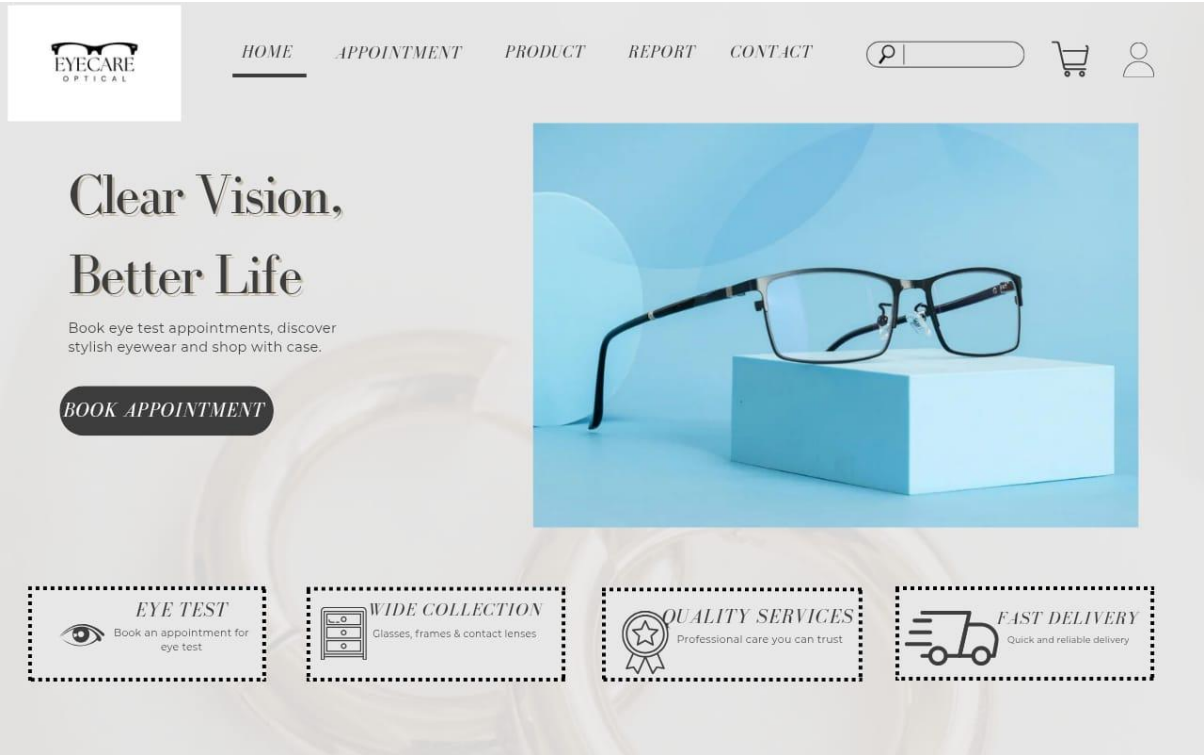


Figure 3- Home page UI

2. Login Page

The Login Page allows registered users such as customers, staff members, and administrators to securely access the system using their email and password. The page includes authentication features and error validation for incorrect login credentials.

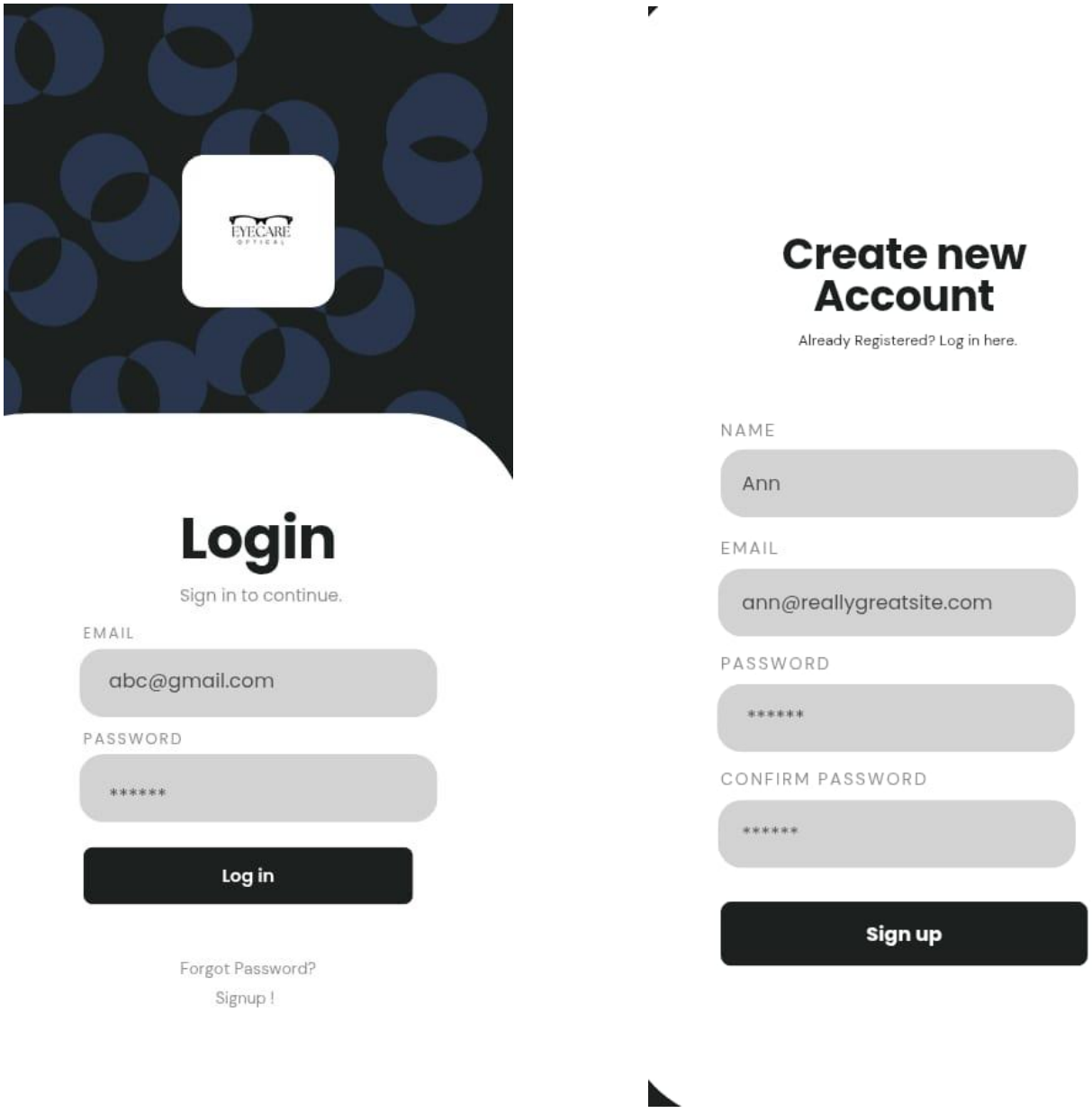


Figure 4- Login page UI

3. Shopping Cart Page

The Shopping Cart Page allows customers to view selected optical products before completing a purchase. Users can update product quantities, remove items, view total prices, and proceed to the checkout process for payment confirmation.

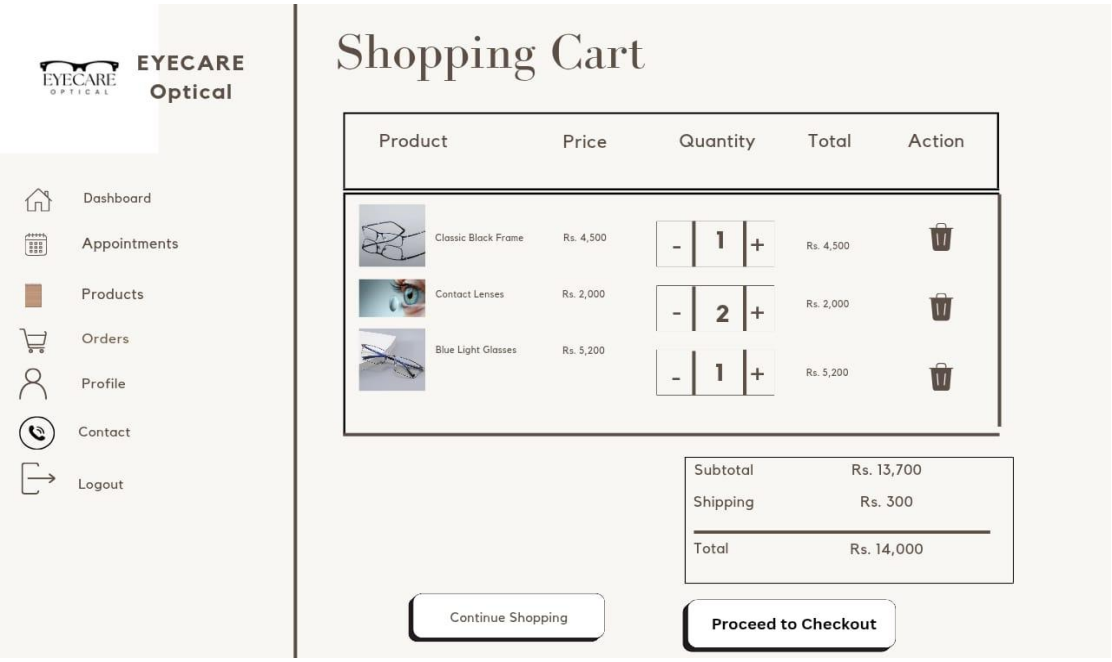


Figure 5- Shopping cart page UI

4. Order Tracking Page

The Order Tracking Page enables customers to monitor the status of their product orders in real time. Customers can view order details, shipping progress, estimated delivery dates, and payment status updates.



Figure 6- Order tracking page UI

5. Add Product Page

The Add Product Page allows administrators to add new optical products into the system. Admin users can enter product details such as product name, category, description, price, stock quantity, and upload product images.



Figure 6- Add product page UI

6. Appointment Page

The Appointment Page allows customers to book eye examination appointments online. Users can select available dates and times, confirm appointments, and receive booking confirmation messages through the system.

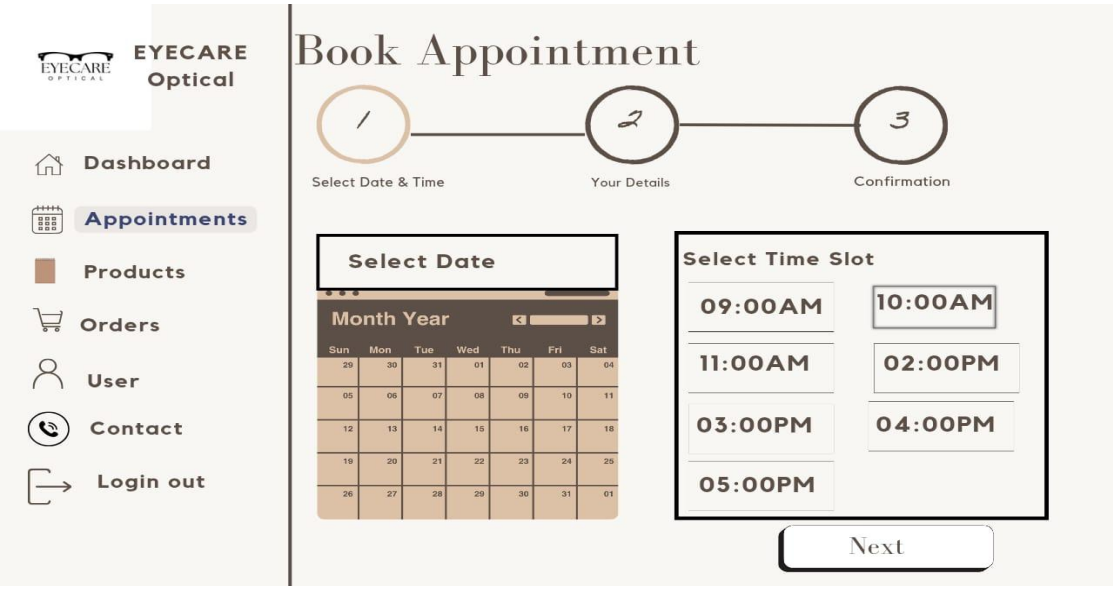


Figure 6- Appointment page UI

7. Product Page

The Product Page displays all available optical products including glasses, frames, and contact lenses. Customers can search products, view product details, compare prices, and add products to the shopping cart.

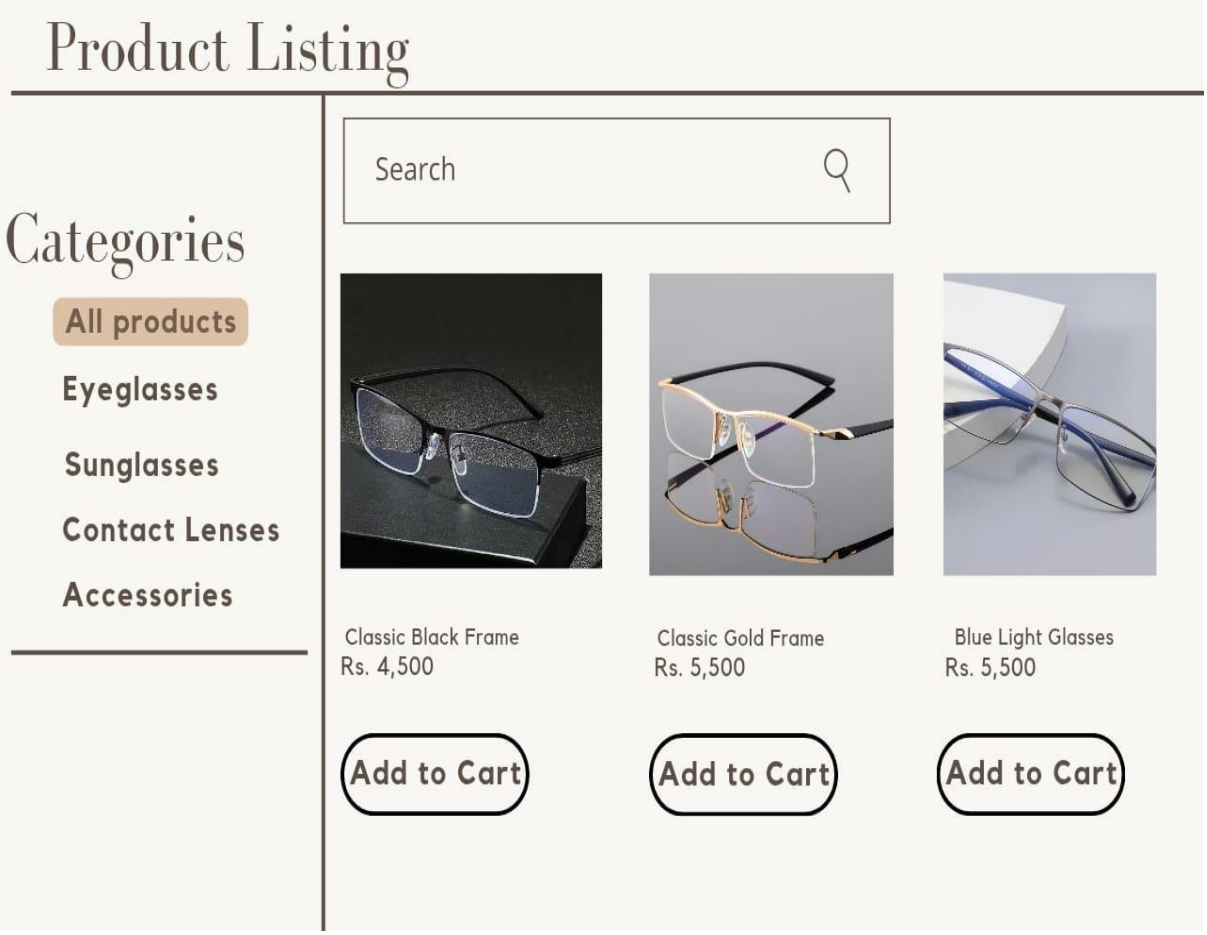


Figure 7- Product page UI

8. Admin Dashboard Page

The Admin Dashboard Page provides administrators with access to manage system operations. Admin users can monitor appointments, manage products, track sales, manage users, generate reports, and control inventory through this centralized dashboard.



Figure 8- Admin dashboard page UI

8.2 Wireframes

Wireframes are designed for:

1. Home Page Wireframe

The Home Page wireframe represents the basic layout structure of the homepage including the navigation bar, banner section, featured products, services section, and footer. It focuses on content placement and user navigation flow.

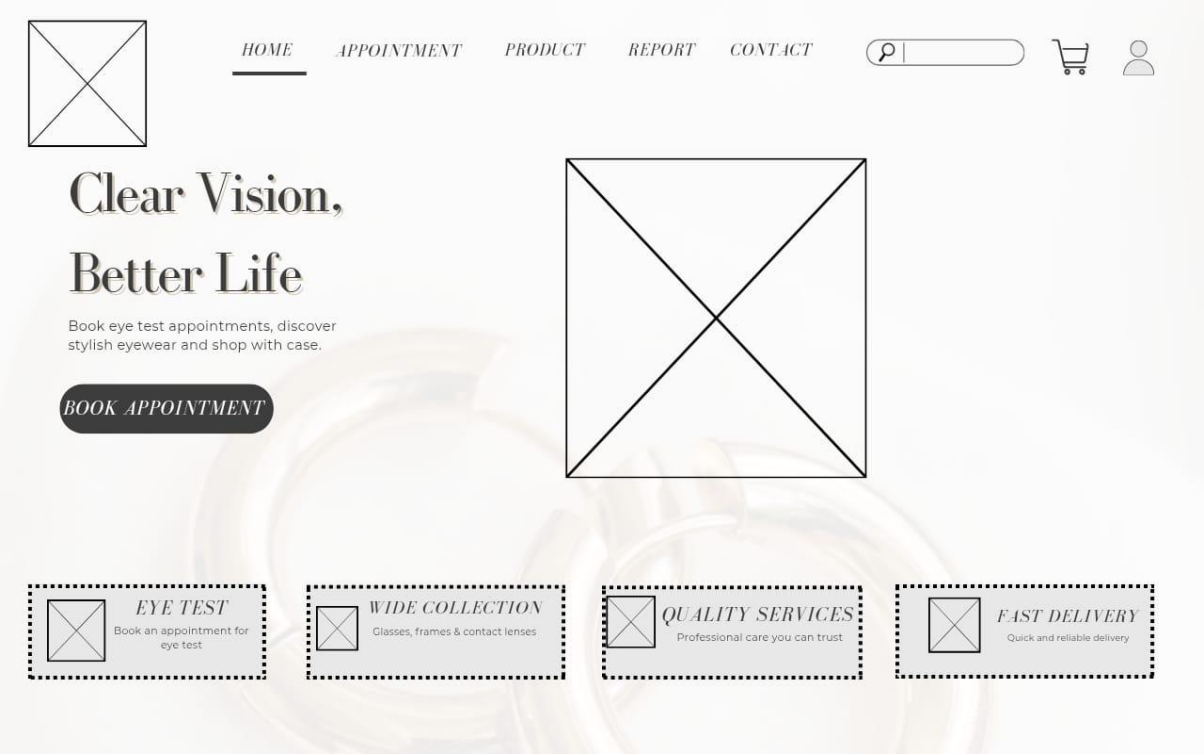


Figure 1- Home page Wireframe

2. Login Page Wireframe

The Login Page wireframe illustrates the placement of login form elements such as email input, password field, login button, and registration link. It ensures a simple and user-friendly authentication interface.

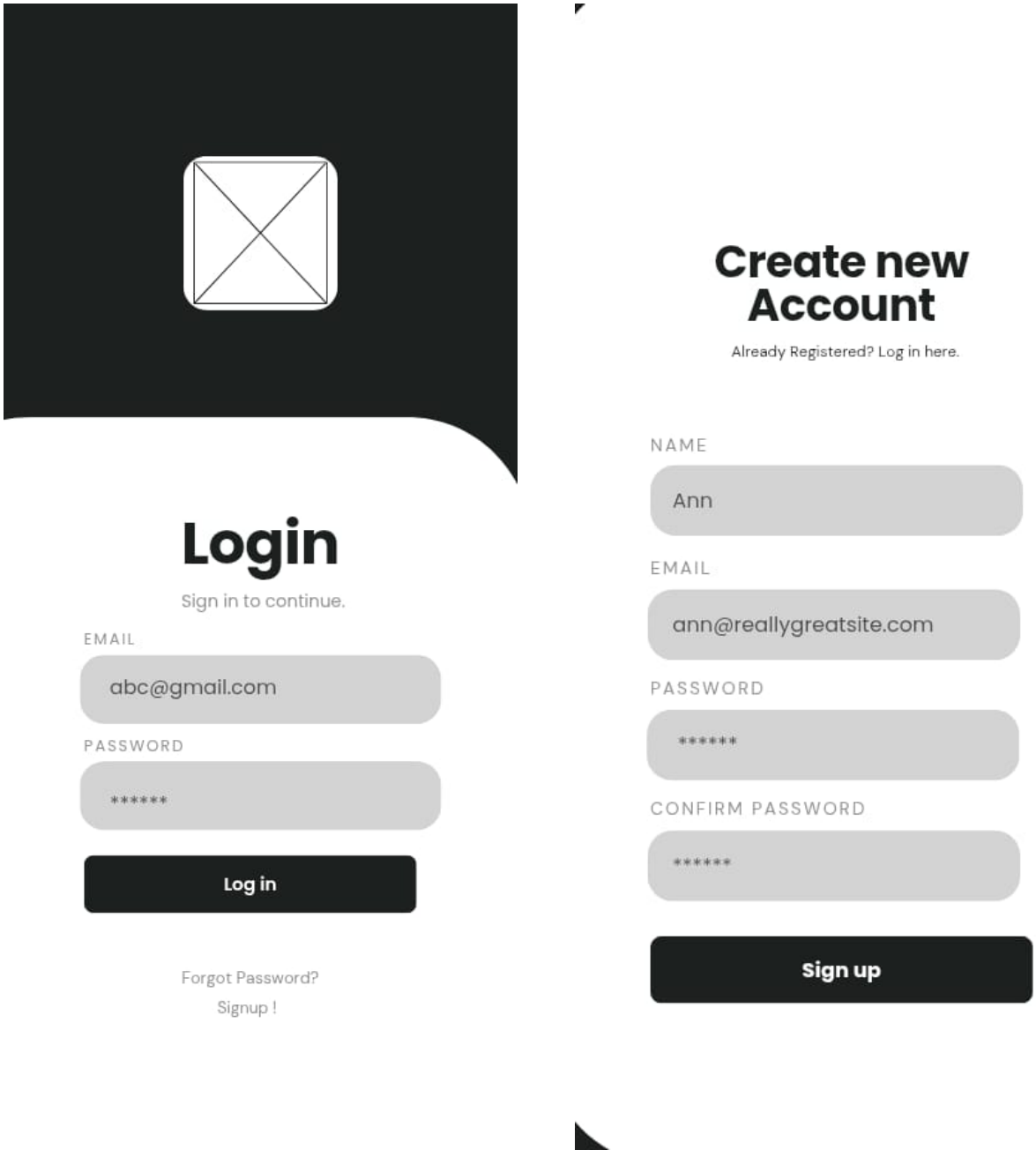


Figure 2- Login page Wireframe

3. Shopping Cart Wireframe

The Shopping Cart wireframe displays the structure of cart items, quantity controls, product prices, total amount section, and checkout button. It helps organize the shopping experience clearly.



Figure 3- Shopping Cart Wireframe

4. Order Tracking Wireframe

The Order Tracking wireframe outlines the layout for displaying order details, tracking status, shipping progress, and delivery information in a clear and organized format.

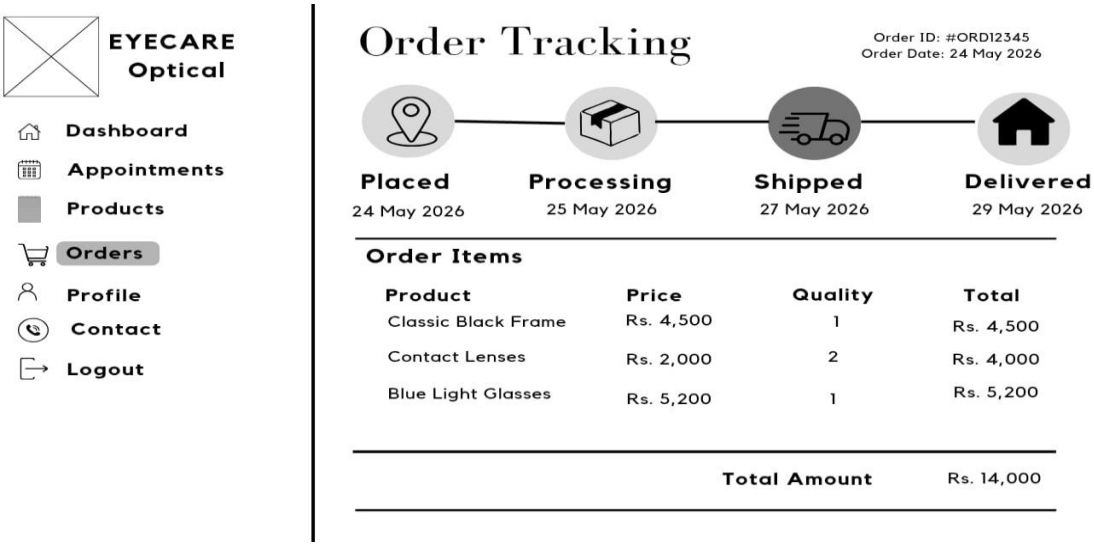


Figure 4- Order Tracking Wireframe

5. Add Product Wireframe

The Add Product wireframe demonstrates the arrangement of product input fields including product name, description, category, price, stock quantity, and image upload section for administrators.

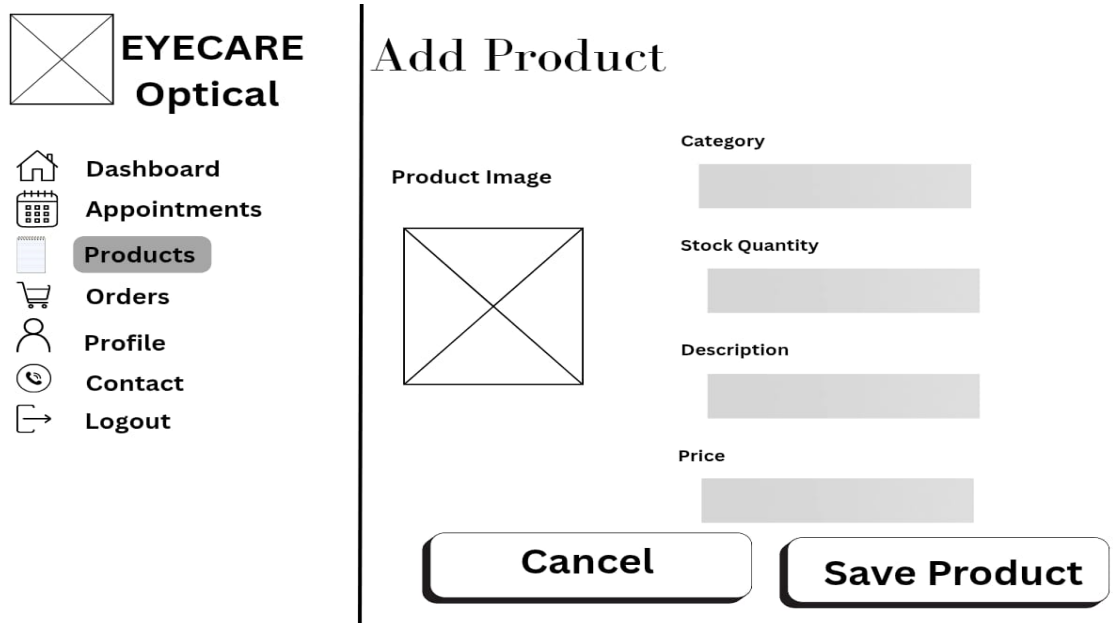


Figure 5- Add Product Wireframe

6. Appointment Page Wireframe

The Appointment Page wireframe presents the structure of appointment scheduling components such as date selection, time slots, customer details form, and booking confirmation button.

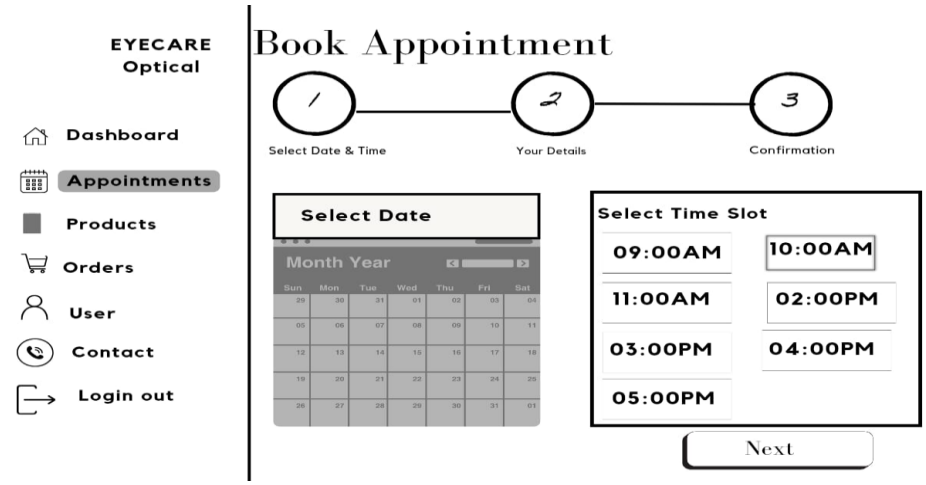


Figure 6- Appointment Page Wireframe

7. Product Page Wireframe

The Product Page wireframe shows the placement of product cards, search bar, category filters, product images, pricing details, and add-to-cart buttons for customer interaction.

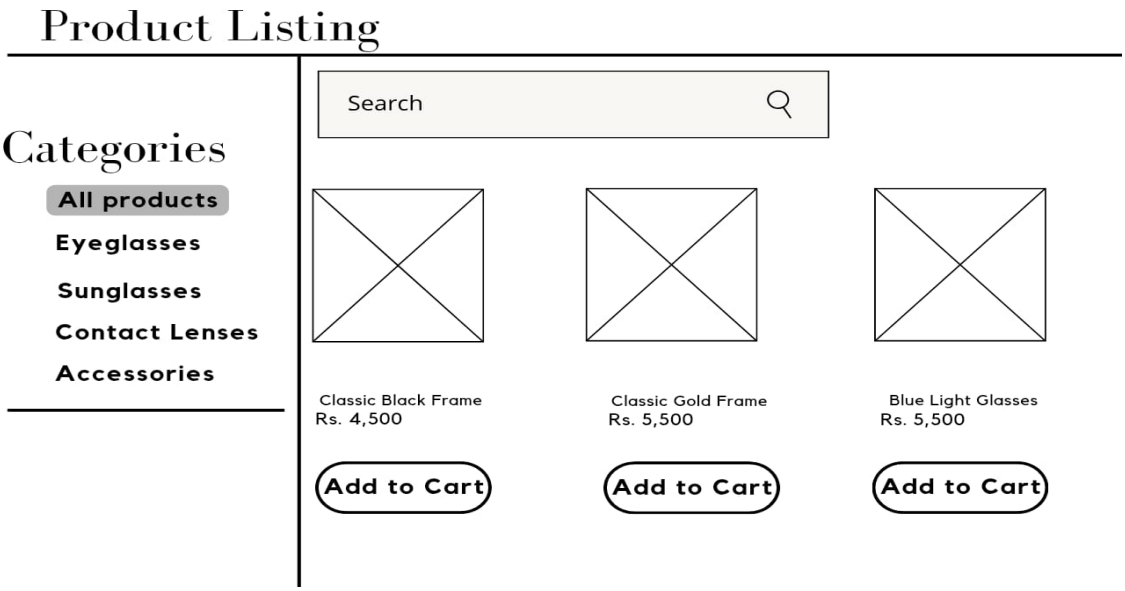


Figure 7- Product Page Wireframe

8. Admin Dashboard Wireframe

The Admin Dashboard wireframe illustrates the layout of the admin control panel including side navigation menus, report summaries, charts, appointment statistics, product management sections, and quick action buttons.



Figure 8- Admin Dashboard Wireframe

9. Process Design

9.1 System Workflow

- Customer Registration
- User Authentication
- Appointment Booking/Product Ordering
- Prescription Management
- Payment Processing
- Report Generation

9.2 Flowcharts

Flowcharts include:

- Login process
- Booking process
- Order process
- Payment process

9.3 Data Flow Diagram

Processes:

- User Input
- Data Validation
- Database Processing
- Output Generation

10. Security Design

10.1 Authentication Mechanism

- JWT Authentication
- Encrypted Passwords
- Session Management

10.2 Data Security

- Role-based access control
- Database backup
- Input validation

10.3 Communication Security

- HTTPS communication
- Secure API requests
- Secure payment gateway integration

11. Error Handling

11.1 Error Detection

- ✓ Input validation
- ✓ Database exception handling
- ✓ Authentication validation

11.2 Error Recovery

- ✓ User-friendly error messages
- ✓ Retry mechanisms
- ✓ Database recovery support

12. Testing Strategy

12.1 Unit Testing

Each module is tested independently.

12.2 Integration Testing

All integrated modules are tested together.

12.3 System Testing

Complete system testing ensures all requirements are satisfied.

13. Appendices

Appendix A – Glossary

Contains definitions of technical terms used in the document.

Appendix B – Sample Interfaces

Includes screenshots/wireframes of:

- Homepage
- Login page
- Product page
- Appointment page
- Admin dashboard

Appendix C – Future Enhancements

AI Prescription Recommendation

AI-based lens and frame recommendations.

Mobile Application

Android and iOS mobile application support.

SMS Notifications

Appointment reminders via SMS.

Online Video Consultation

Virtual eye consultation system.